Extended Data Fig. 5 | High- and low-dimensional velocity evaluation

a, Original-style scatter plots separated by the 14 dyngen topologies. Each point summarizes one method across available matched seed datasets (mean and sample SD); high-dimensional and PCA2 scores are cell-level cosine similarities. b, Paired high/low-dimensional violin plots, faceted by method; lines join the same dataset, with dataset medians shown as points. c, Cosine similarity between predicted and simulated true RNA velocity in PCA spaces of 2–30 dimensions, restoring the original accuracy-curve definition. Each colored point is the median of finite method–dataset cell-level cosine scores within one topology at that dimension; the black total curve is the median across all 42 datasets and available methods. These are projected velocity-accuracy scores, distinct from the high/low-dimensional score-vector consistency in Fig. 3d. Seventeen available configurations are included; missing scores are excluded without imputation. Source scores and valid counts are supplied. TopicVelo table values retain unverified native-transition provenance. No model was rerun.